



ELE1204: Digital Electronics and Logic Design

Lecture 08 (BSCS)

Programmable Logic Devices

Ms. Olivia Nakayima

(MSc, BSc. CCNA, JNCIA, JNCIS)

Department of Computing & Technology

Faculty of Engineering, Design & Technology

Mon 16th October 2023

A Complete Education for A Complete Person

P.O. Box 4, Mukono, Uganda, Plot 67-173, Bishop Tucker Road, Mukono Hill | Tel: +256 (0) 312 350 800 Email: info@ucu.ac.ug Web: <https://ucu.ac.ug>

Founded by the Province of the Church of Uganda. Chartered by the Government of Uganda



Lecture Objectives and Learning outcomes

The Objectives of this lecture are :

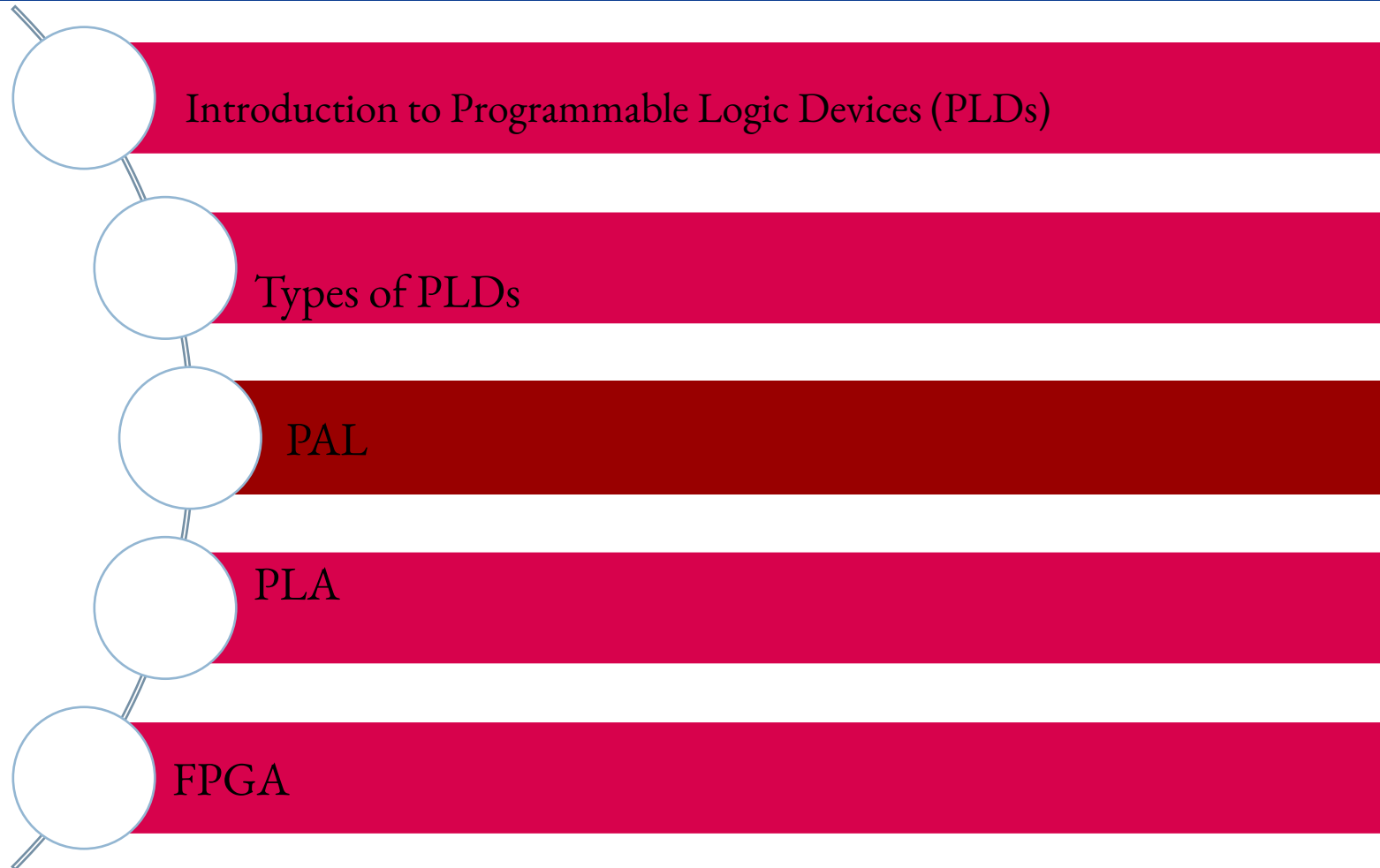
- ☐ To understand PLDS
- ☐ To compare and contrast their different types.

NOTE:: SMART = **S**pecific, **M**easurable, **A**chievable, **R**elevant, **T**imebound





Lecture Overview



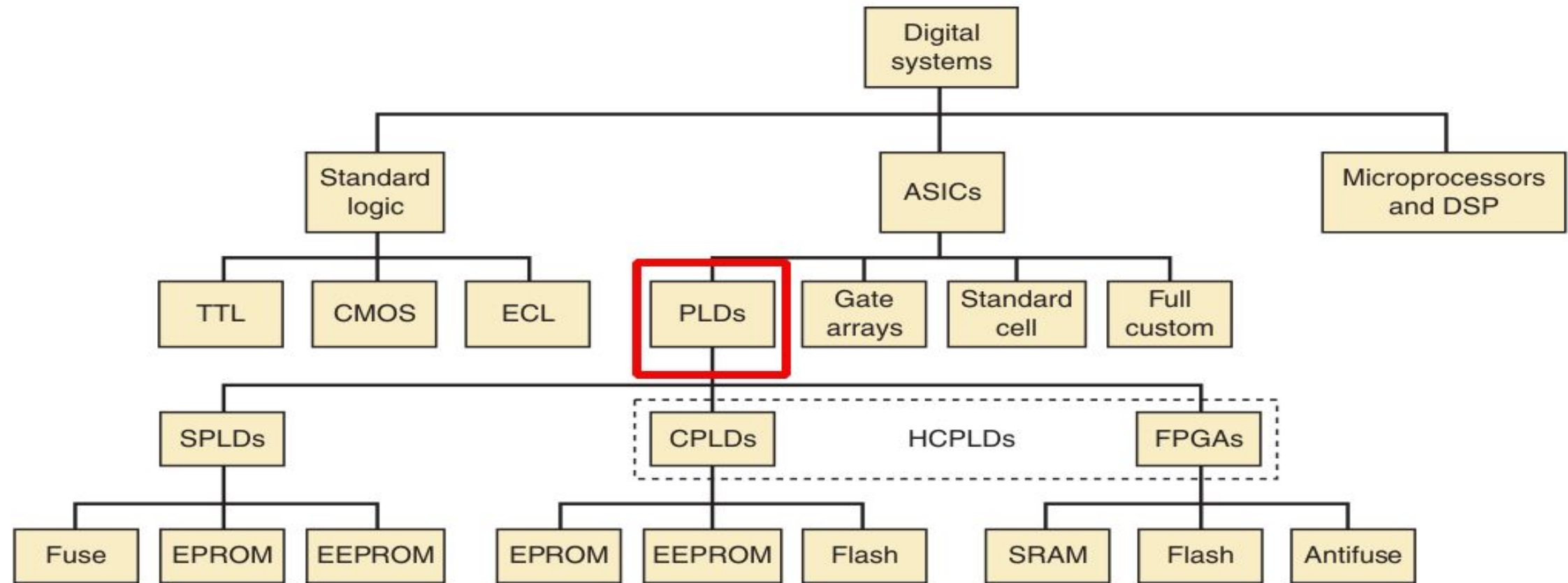


FIGURE 13-1 Digital system family tree.



Introduction to Programmable Logic Devices (PLDs)

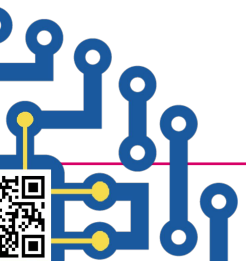
Programmable Logic Devices (PLDs) are digital circuits where the logic configuration can be **programmed after manufacturing**, allowing flexibility in digital system design.

Original devices could be programmed only once and, if the design changed, the old PLD would have to be removed from the circuit and a new one, programmed with the updated design, would have to be inserted in its place. With newer devices, the internal logic design can be changed on the fly, while the chip is still connected to a printed circuit board in an electronic system.

Types of PLDs

PLDs are classified into:

1. **Simple PLDs (SPLDs)** – Includes **Programmable Logic Array (PLA)** and **Programmable Array Logic (PAL)**.
2. **Complex PLDs (CPLDs)** – Combine multiple SPLDs with advanced features.
3. **Field-Programmable Gate Arrays (FPGAs)** – Highly flexible and reconfigurable.





PAL (Programmable Array Logic)

This is a programmable logic device with a **fixed OR array** and a **programmable AND array**.

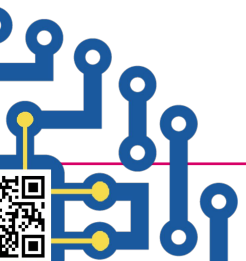
https://www.youtube.com/watch?v=qlq4NHk5Y_w

Structure:

- **AND Array (Programmable):** Allows customization of input logic.
- **OR Array (Fixed):** Determines the final logic output.

Advantages of PAL:

- ✓ Faster than PLA because it has **fixed OR gates**.
- ✓ Simple and cost-effective for small logic designs.
- ✓ Easy to program and modify.





PLA (Programmable Logic Array)

PLA is a programmable logic device with **both AND and OR arrays programmable**, offering more flexibility than PAL.

Structure:

- **AND Array (Programmable):** Can implement any number of minterms.
- **OR Array (Programmable):** Can sum any combination of minterms.

Advantages of PLA:

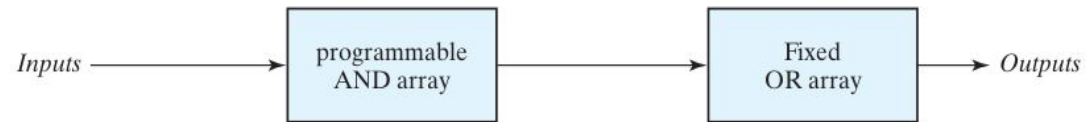
More flexible than PAL.

Can implement **multiple complex logic functions**.

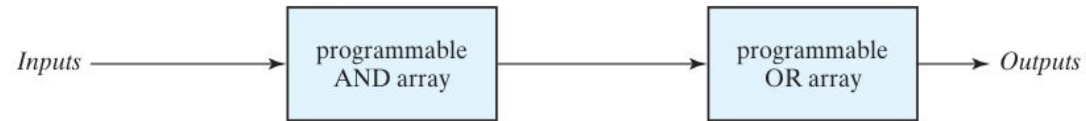
Used in applications where logic needs frequent updates.



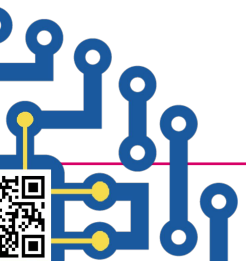
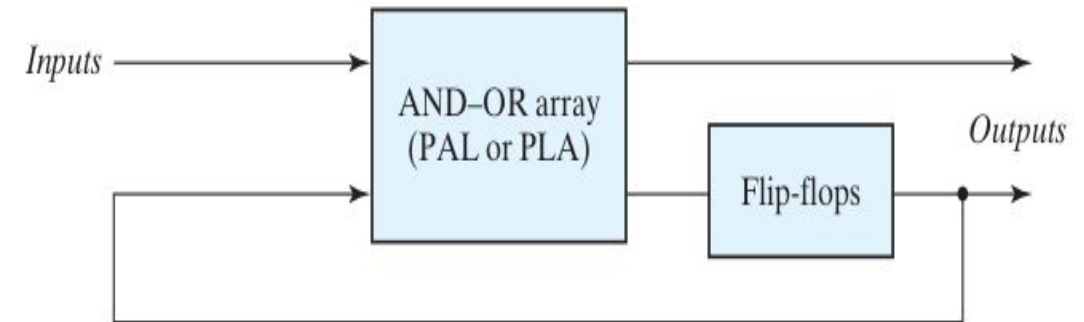
PLA and PAL



(b) Programmable array logic (PAL)



(c) Programmable logic array (PLA)





FPGA (Field-Programmable Gate Array)

FPGA is an advanced PLD that contains a **large number of logic gates, flip-flops, and routing paths** that can be configured after manufacturing. <https://www.youtube.com/watch?v=WY-F3knh7c>

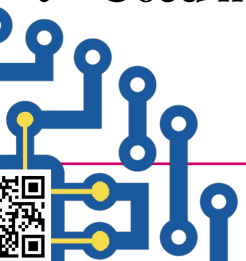
It can be reprogrammed using (Hardware description Languages)HDL languages (Verilog,)

Structure:

- **Configurable Logic Blocks (CLBs):** Contain logic functions and flip-flops.
- **Interconnect Matrix:** Programmable connections between logic blocks.
- **Input/Output Blocks (IOBs):** Handle external communication.

Advantages of FPGA:

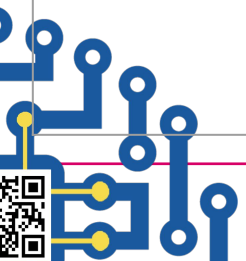
- ✓ **Highly flexible** and can be reprogrammed.
- ✓ **Parallel processing** capability, making it ideal for high-speed applications.
- ✓ Used in AI, networking, automotive, and aerospace applications.





Comparison of PAL, PLA, and FPGA

Feature	PAL	PLA	FPGA
Programmable AND Gates	Yes	yes	yes
Programmable OR Gates	fixed	yes	yes
Flexibility	Medium	high	Very high
Complexity	Low	Medium	High
Reconfigurability	One time programmable	no	yes
Speed	Fast	Slower than PAL	High speed
Common use	Simple logic designs	Medium complexity circuits	High-performance application(signal processing, AI)





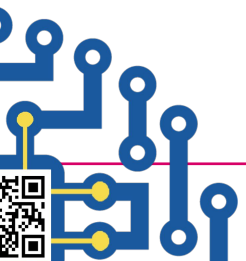
Applications of PLDs

Embedded Systems: Used in microcontrollers and control systems.

✓ **Networking:** FPGA-based network routers and switches.

✓ **AI & Machine Learning:** FPGA accelerators for deep learning models.

✓ **Aerospace & Automotive:** Safety-critical systems where reconfigurable logic is needed.





UGANDA CHRISTIAN
UNIVERSITY

A Centre of Excellence in the Heart of Africa



Uganda Christian University

P.O. Box 4 Mukono, Uganda

Tel: 256-312-350800

 <https://ucu.ac.ug/> Email: info@ucu.ac.ug.

 @ugandachristianuniversity  @UCUniversity

 @UgandaChristianUniversity



Department of Computing & Technology FACULTY OF ENGINEERING, DESIGN AND TECHNOLOGY

Tel: +256 (0) 312 350 863 | WhatsApp: +256 (0) 708 114 300

 @ucuc Computeng  @ucu_ComputEng

 <https://cse.ucu.ac.ug/> Email: dct-info@ucu.ac.ug

A Complete Education for A Complete Person

P.O. Box 4, Mukono, Uganda, Plot 67-173, Bishop Tucker Road, Mukono Hill | Tel: +256 (0) 312 350 800 Email: info@ucu.ac.ug Web: <https://ucu.ac.ug>

Founded by the Province of the Church of Uganda. Chartered by the Government of Uganda